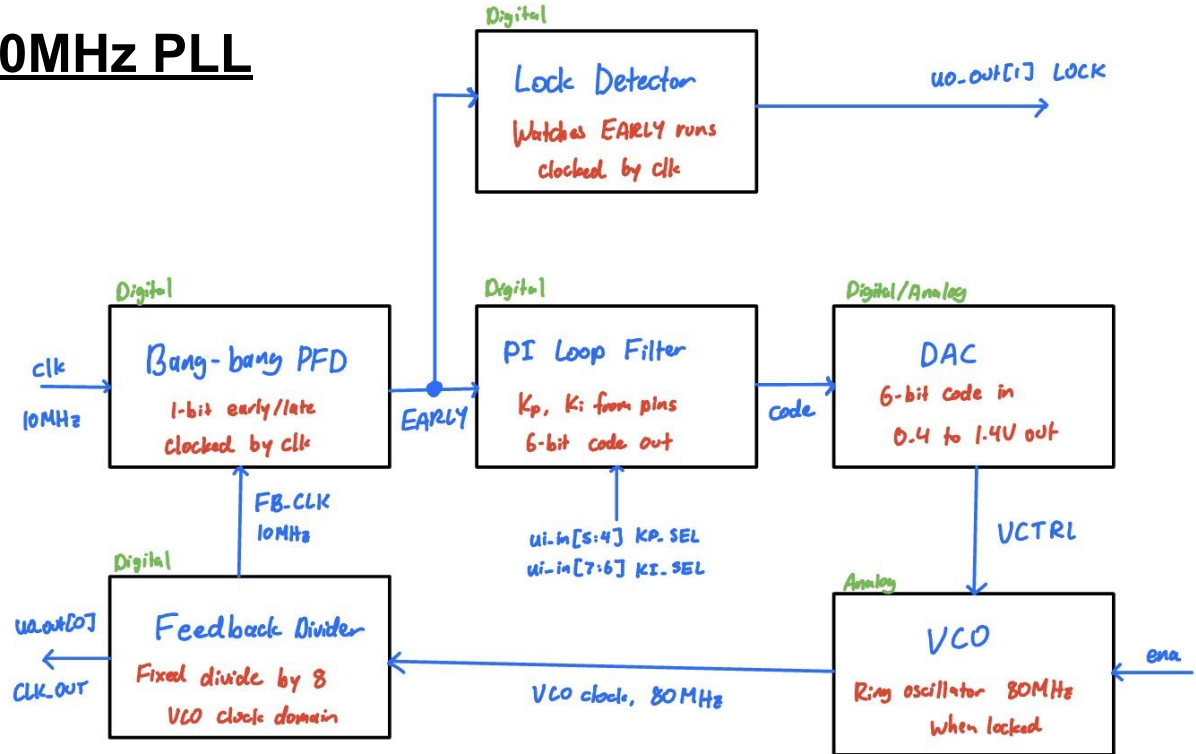


Bang-Bang 10MHz $\rightarrow$ 80MHz PLL

Statement of Purpose

We will build a Bang-Bang phase-locked loop that multiplies a 10 MHz reference clock up to 80 MHz.

It pairs a PI filter with a voltage-controlled oscillator (VCO), with the input fed by a DAC



Category	Spec	Notes
Supply	1.8 V	Analog current under 1 mA
Reference frequency	10 MHz	From clk
Multiplication ratio	N = 8, fixed	Hard-wired in the feedback divider
Output frequency	80 MHz	12.5 ns period
Phase detector	Bang-bang phase-frequency detector	1 decision per reference cycle
Loop filter	Digital PI	Kp and Ki each have 4 pin-selected settings
DAC	6-bit, monotonic, output 0.4 to 1.4V	About 16 mV per code

VCO tuning range	50 to 110 MHz at tt over the DAC range	80 MHz must stay inside the range at ss and ff corners. Gain near 60 MHz/V
Frequency step	About 1 MHz per code	DAC step × VCO gain, about 1.2% of 80 MHz
Frequency acquisition	Locks from mid-code to 80 MHz at every corner	No manual help
Lock time	Under 1000 reference cycles	100 μs
Locked dither	Within ±2 codes	Sets output jitter. Reported from simulation
Lock detector	Asserts after 256 cycles with no run of more than 8 identical decisions	Thresholds tuned in simulation
Modes	Closed loop, open loop	Open loop sets the DAC code from pins and exposes VCTRL on ua[0]
Output pins	10 MHz on uo_out[0] at the default divide	VCO divided by 8 or 32 before it leaves the chip

Member	Owns
Benjamin DONG	Bang-bang phase frequency detector
Gabe HOH	Feedback divider, PI filter
Hugh DING	Lock Detector + DAC, Integration
Matthew CHEN (CAPTAIN)	Voltage-Controlled Oscillator, Top level sim
All	Tool setup, Python/Spice simulations, final report

